

Nessma Mohdy

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EDUCATION

Bachelor of Engineering in Software and Mechatronics Engineering @ University of Calgary

- Key Courses: Embedded & Hardware Systems, Networking, Data Structures & Algorithms, Linear Algebra, Applied Operating Systems, Machine Learning, Digital Circuits, Control Systems, Computer Architecture, Software Testing, VR

EXPERIENCE

Firmware Intern @ AMD

May 2025 - Present

- Applied C programming skills to work on OpenSIL, AMD's open-source silicon initialization firmware, gaining exposure to hardware-software interfaces and firmware development processes.
- Utilizing Coverity and CERT to identify, analyze, and remediate security vulnerabilities and code quality issues in low-level firmware, reinforcing best practices in secure and reliable system development.

Machine Vision Software Engineer @ Tesla

January 2025 – May 2025

- Increased team efficiency by creating a web platform (Flask, InfluxDB, Caspian) that visualizes cell data on a heatmap and integrates Pytorch/Halcon for advanced analysis navigating millions of images.
- Implemented a comprehensive vision system (camera selection, setup, installation) utilizing ResNet for defect detection and machine vision for crack identification 94+% detection accuracy.
- Improved OCR system for cell ID recognition, achieving <0.2s average recognition time per unit and increasing reliability to 98.7%.

Controls Engineering Team Member @ Schulich Space Rover

October 2023 – December 2024

- Migrated Python-based rover control and inter-module WebSocket communication to C++ for substantial performance gains and improved system-level execution.
- Executed PID control algorithms to regulate motor speed, enhancing precision and stability in control systems through feedback loops and OP-AMPS.

Haptic Robotic Arm & Data Analyst Researcher @ University of Calgary

May – August 2023

- Optimized non-linear supervised machine learning methods in MATLAB with an analytic mindset that aided in the publication of a research paper on long COVID effects resulting in a 83% accuracy score.
- Guided a team that oversaw PCB design, 3D printing of the open-source THOR robot on Autodesk Fusion, assembled around the inverse kinematics, and dealt with soldering.

PROJECTS

LED Controller | C

- Based on datasheets and hardware schematics I engineered an embedded pulse width modulation system for LED lights, programmed for a PIC24 micro-controller. Led circuit design responsive to buttons through interrupts. Utilized Spyder's IDE for Python and UART communications to graphically display the light intensity.

Scavenger Hunt Video Game | C#, Unity

- Integrated haptic feedback into a Unity 3D multi-level game (C#), enhancing player immersion.

AI Obituary | React, AWS

- Innovated a React Full Stack web application taking note of the software cycle, enabling users to create their own collection of GPT-4 RESTful API generated obituaries. Developed a terraform configuration to connect to AWS Amazon Polly, lambda functions, dynamoDB to ensure an optimal user experience.

Optimal Sums | C, Linux

- Developed sequential and parallel implementations of array sum using Hillis-Steele algorithm, with configurable threading for optimized performance.

LongTimeNoCrypto | C++, Networking

- Modified the open source sha256 encryption and created a Cipher Hill cryptic algorithm to create respective hash blocks. Utilized problem-solving skills to create a P2P system that sends a secrete message, added to a blockchain

TECHNICAL SKILLS

Coding Languages: C/C++, Python, Java, SQL, Arduino, JavaScript, HTML, MATLAB, C#, ROS, RISCv Assembly

Platforms: Linux, Windows, Mac OS

Tools: Git, CAD, Unity, VS Code, Blender, MySQL, Quartus, MPLAB X, XML, UTM, TCP communication, Microcontrollers, RTOS, UART, GUI, Jira, Docker, Terraform, Ansible, Kubernetes, Excel, Agile, Eclipse, Simulink